“The vast majority of large mutations are deleterious; small mutations are both far more frequent and more likely to be useful.”

Ronald Fisher

Different rates of evolution

The ancestors of all life on Earth were very simple organisms. Recent scientific research suggest that the earliest “biogenic” rocks—derived from early life forms—date back nearly four billion years. In that time, highly complex life forms have evolved, and later fossils of species that look more similar to those of today reveal what has occurred. For example, a fossil record stretches back 60 million years for ancestors of the horse. The earliest of these were dog-sized forest-dwelling animals with several toes on each foot. Evolution produced much larger horses with just a single hoof on each foot, adapted for life on open grasslands where they would often have had to outrun predators.

Peppered moths (*Biston betularia*) reveal change over a shorter period. The moth is usually pale, providing camouflage against the bark of birch trees, but a mutation produces some black moths. Before the 19th century, most peppered moths were pale. During the Industrial Revolution (1760–1840), however, smoky air left deposits of soot on trees and buildings in British cities, and the black form became much commoner. By 1895, 95 percent of peppered moths in Britain’s cities were black, as paler moths were eaten by birds because their coloring provided no camouflage. This phenomenon continues to act as an example of Darwin’s theory in action today, as the pale moth becomes common once more due to the declining soot concentrations in Britain’s cities.